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PLAINQAFACT: Retrieval-augmented Factual Consistency Evaluation Metric for Biomedical Plain Language Summarization

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Abstract

Hallucinated outputs from large language models (LLMs) pose risks in the medical domain, especially for lay audiences making health-related decisions. Existing automatic factual consistency evaluation methods, such as entailment- and question-answering (QA) -based, struggle with plain language summarization (PLS) due to *elaborative explanation* phenomenon, which introduces external content (e.g., definitions, background, examples) absent from the scientific abstract to enhance comprehension. To address this, we introduce PLAINQAFACT, an automatic factual consistency evaluation metric trained on a fine-grained, human-annotated dataset PLAINFACT, for evaluating factual consistency of both source-simplified and elaborately explained sentences. PLAINQAFACT first classifies sentence type, then applies a retrieval-augmented QA scoring method. Empirical results show that existing evaluation metrics fail to evaluate the factual consistency in PLS, especially for elaborative explanations, whereas PLAINQAFACT consistently outperforms them across all evaluation settings. We further analyze PLAINQAFACT's effectiveness across external knowledge sources, answer extraction strategies, answer overlap measures, and document granularity levels, refining its overall factual consistency assessment. Taken together, our work presents a sentence-aware, retrieval-augmented metric targeted at elaborative explanations in biomedical PLS tasks, providing the community with both a robust benchmark and a practical tool to advance reliable and safe plain language communication in the medical domain. PLAINQAFACT and PLAINFACT are available at: `https://github.com/zhiwenyou103/PlainQAFact`.

Keywords: Plain language summarization, Factual consistency evaluation, Retrieval-augmented generation, Hallucination, Large language models

1. Introduction

Communicating biomedical scientific knowledge in plain language is essential for improving health information accessibility and health literacy [1, 2]. Recent advances in large language models (LLMs) have made significant progress in plain language summarization (PLS) of biomedical texts [3, 4, 5, 6]. However, ensuring the factual consistency of these summaries remains a major challenge. A key source of inconsistency stems from *elaborative explanations*: content such as definitions, background information, and examples that enhance comprehension but are not explicitly present in the original scientific abstracts (i.e., source) [7, 8, 9]. While such elaborations are critical for effective communication, they introduce external content that cannot be directly verified against the source, complicating automatic factual consistency evaluation.

Factual consistency in PLS is typically assessed through a combination of human evaluation and automated metrics [3, 10]. While human evaluation is reliable [11], it is costly and difficult to scale, particularly in biomedical contexts where domain expertise is required. Commonly used factuality metrics can effectively verify content supported by the source but fail to assess factual consistency of added information [12]. However, these metrics depend heavily on high-quality reference summaries, which are often unavailable in plain language summaries. Recent prompt-based evaluation techniques show promise [13, 14], but their sensitivity to factual perturbations in elaborative content remains limited [12].

The lack of suitable benchmark datasets further hinders progress. Many existing datasets are constructed from LLM-generated summaries or apply rule-based perturbations to simulate non-factual content. For example, FactPICO provides expert annotations for plain language summaries of randomized controlled trial abstracts, focusing on PICO elements and evidence inference [9]. However, it includes factuality labels only for added content, leaving simplified sentences unannotated, which are generated by LLMs and potentially inaccurate. In contrast, APPLS perturbs human-written summaries using rule-based transformations [12], but cannot ensure that the resulting outputs remain coherent or factually plausible. These limitations underscore the need for high-quality, sentence-level annotations grounded in human-authored summaries.

To address these challenges, we propose a targeted retrieval-based metric for factual consistency evaluation in PLS (Figure 1). We introduce a new expert-annotated dataset, PLAINFACT, consisting of human-written plain language summaries aligned with scientific abstracts. Each sentence is labeled with its type (elaborative explanation vs. source simplification), functional role, and alignment to the source (§3.1). Building on this and prior studies for the QA-based evaluation metrics in the general

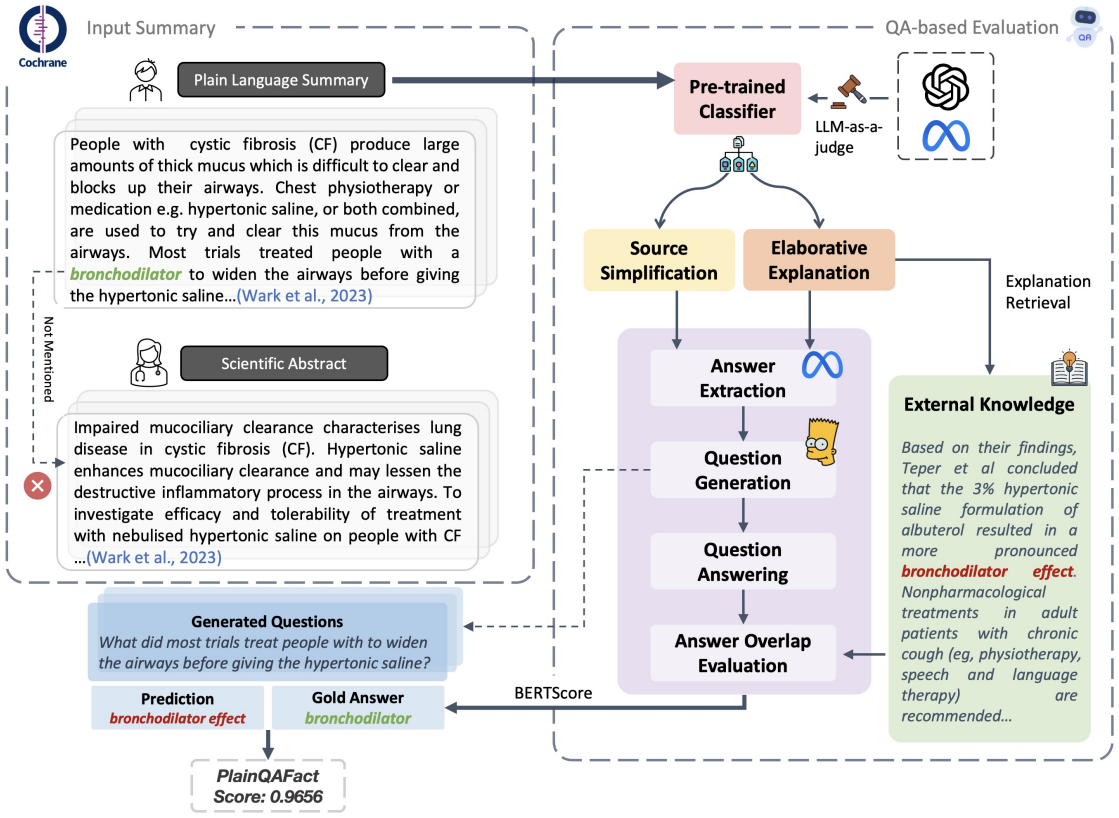


Figure 1: Overview of PLAINQAFACT. A fine-tuned classifier first identifies the sentence type, involving either source simplification or elaborative explanation. Then, a QA-based evaluation pipeline performs answer extraction, question generation, question answering, and answer overlap evaluation. For elaborative content not present in the scientific abstract, PLAINQAFACT retrieves external knowledge to verify factual consistency. The illustrated example shows an elaborative explanation involving a "bronchodilator" not mentioned in the source abstract but verifiable through external evidence. PLAINQAFACT assigns a high score, reflecting strong alignment between the extracted and gold answers.

domain [20, 21], we present PLAINQAFACT, a dual-stage QA-based evaluation metric that selectively applies retrieval only to elaborative explanations (§3.2). Experiment results on several PLS datasets demonstrate the effectiveness of our approach, particularly in assessing complex, elaborative content (§4, §5). Taken together, our work presents the first metric specifically designed to evaluate the factual consistency in elaborative explanations in PLS, providing the community with both a robust benchmark and a practical tool to advance reliable and safe plain language communication in the medical domain.

2. Related Work

Limitations of Existing Factuality Evaluation. The primary approach for evaluating plain language generation combines automated metrics with human evaluation [3, 10]. While human assessment provides a thorough analysis [11], its high cost and time demands make it impractical for large-scale datasets. Evaluating factual consistency in biomedical PLS is particularly challenging, as it requires domain expertise. Entailment- [15, 16], similarity- [17, 18], model- [19] and QA-based [20, 21] metrics are commonly used for factual consistency assessment but rely heavily on high-quality reference summaries, which are often unavailable or difficult to obtain for PLS. Recent advancements in prompt-based evaluation show promise [13]; however, their sensitivity to factuality perturbations in PLS remains limited [12]. To address these limitations, we propose a reference-free solution for factual consistency evaluation of PLS that effectively assesses factual consistency with external information retrieval to augment the reference summary.

Retrieval-Augmented Generation. Retrieval-augmented methods enhance text generation by extracting relevant information from external sources to supplement input queries [22]. These methods have been shown to be effective in open-domain QA [23, 24], knowledge-based QA [25], and multi-step reasoning [26, 27]. In the context of PLS, retrieval from structured knowledge bases (KBs) has been shown to improve factual accuracy compared to language models alone [28]. However, retrieval-augmented approaches have not been extensively explored for factual consistency evaluation in PLS, despite their potential for addressing elaborative explanations. In this work, we investigate retrieval-augmented QA to enhance PLS factual consistency assessment while also examining its limitations.

3. Methods

We first introduce the protocol of PLAINFACT dataset curation (§3.1). Based on this dataset, we propose PLAINQAFACT, a two-stage retrieval-augmented QA framework for factual consistency evaluation in PLS tasks (§3.2).

3.1. PLAINFACT Benchmark

To develop a high-quality factual consistency evaluation benchmark in PLS tasks, we collect a subset from the largest human-authored CELLS [28] dataset (§3.1.1) and hire domain experts to provide fine-grained sentence-level annotations (§3.1.2).

3.1.1. Human-Authored PLS Dataset

Rather than relying on LLM-generated plain language summaries, we construct our benchmark using human-authored summaries. CELLS [28], the largest parallel corpus of scientific abstracts and their corresponding plain language summaries, is written by the original authors and sourced from 12 biomedical journals. We primarily select data from the Cochrane Database of Systematic Reviews (CDSR)[1] within CELLS, as CDSR contains systematic reviews that support evidence-based medical decision-making across healthcare domains [29]. Since systematic reviews represent the highest level of scientific evidence, this selection enhances the factual rigor of our dataset. To ensure readability, we filter the 200 most readable plain language summaries based on average scores from three standard readability metrics: Flesch-Kincaid Grade Level (FKGL) [30], Dale-Chall Readability Score (DCRS) [31], and Coleman-Liau Index (CLI) [32]. Additionally, given the collected plain language summaries from CELLS are all factual, we further conduct sentence-level perturbation to transform each plain language sentence into incorrect ones. Specifically, we use `GPT-4o`[2] to perturb each plain language sentence based on prompts introduced in Guo et al. [12]. Therefore, for PLAINFACT, we have 400 summary-abstract pairs in total, where 200 factual pairs and 200 non-factual pairs.

	Elaborative Explanation	Source Simplification
# of Sentences	1,213	1,527
Average Length (token)	29	28
Vocabulary Size	4,230	4,046
Has Reference	417	1,527
# of Background	533	329
# of Definition	82	44
# of Method/Result	512	1,107
# of Example	10	3
# of Other	76	44
FKGL ↓	12.5	12.4
DCRS ↓	11.3	11.6
CLI ↓	13.5	13.9

Table 1: Overview of the PLAINFACT benchmark. Medical experts annotated 200 pairs of plain language summaries and their corresponding scientific abstracts from the CELLS dataset [28], categorizing each plain language sentence as either a source simplification or an elaborative explanation related to the abstract. Lower scores of FKGL, DCRS, and CLI indicate better readability.

[1] `http:///www.cochranelibrary.com`

[2] We use `gpt-4o-2024-11-20` version for all experiments of `GPT-4o` in this paper.

3.1.2. Expert Annotation

Since each summary-abstract pair is authored by the same individual, we assume all information to be factual. The annotation aims to capture how plain language sentences relate to their scientific abstracts. Annotators analyze each plain language sentence across three dimensions: (1) **Factuality type**: identify whether a sentence is a source simplification (derived from the abstract) or an elaborative explanation (introducing new content); (2) **Functional role**: categorize the sentence as background, definition, example, method/result, or other; and (3) **Sentence alignment**: map each plain language sentence to its corresponding sentence(s) in the scientific abstract. Details of annotation guidelines are provided in Appendix A. Note that "functional role" and "sentence alignment" are not components of developing PlainQAFact. We use these two fields in PlainFact as metadata to conduct error analysis and document our annotation process in Appendix A and Appendix B.

Annotations are conducted by four independent annotators, each with at least a bachelor's degree in biomedical sciences and prior fact-checking experience. Annotators are recruited via Upwork and compensated from $15 to $20 per hour. Each summary-abstract pair is annotated by two independent annotators, with disagreements resolved by a third. Inter-rater agreement (IRA), measured by Cohen's Kappa for factuality type, functional role, and sentence alignment are 0.43, 0.60, and 0.55 respectively computed from 2,740 annotated plain language sentences, indicated fair agreement for all tasks [33].

Table 1 summarizes the dataset characteristics. Notably, 44% of plain language sentences are elaborative explanations, highlighting their role in enhancing the readability of plain language summaries. 66% of these cannot be directly verified against the source abstract, which underscores the need for factuality evaluation methods that account for such phenomena. Moreover, elaborative explanations include more background, definitions, and examples than source simplifications. For annotation examples, see Appendix B. Additionally, we provide qualitative analysis of PlainFact in Appendix C given the fair IRA of factuality type.

3.2. PlainQAFact Framework

PlainQAFact conducts fine-grained factual consistency evaluation for plain language summaries by first segmenting each summary into sentences, classifying as either simplification or explanation, and retrieving tailored external knowledge within a retrieval-augmented QA framework. Figure 1 provides an overview of its three key components: sentence-level classification (§3.2.1), domain knowledge retrieval (§3.2.2), and dual-stage QA-based factual consistency evaluation (§3.2.3).

3.2.1. Learned Factuality Type Classifier

As elaborative explanations are prevalent in plain language generation and existing metrics struggle to capture added information [12], we first fine-tune a pre-trained language model to categorize factuality types of plain language sentences. Based on the factuality type annotations (source simplification vs. elaborative explanation) in PLAINFACT, we fine-tune the `PubMedBERT-base` model[3] as a classifier. Our PLAINFACT is split 8:1:1 for training, validation, and testing. Additionally, we compare the pre-trained classifier with GPT-4o [34] as a zero-shot classifier to explore extending our factual consistency evaluation metric to domains and tasks lacking human-annotated data.

3.2.2. Domain Knowledge Retrieval

Retrieval-augmented methods have proven effective for explanation generation [28, 35], making them a natural fit for evaluating elaborative explanations. Since our dataset is in the medical domain, we employ MedCPT [36], a retriever pre-trained on large-scale PubMed search logs to generate biomedical text embeddings. For external courpus, we incorporate StatPearls[4] [37] for clinical decision support and medical Textbooks[5] [38] for domain-specific medical knowledge (see details in Appendix D.

3.2.3. QA Evaluation Components

QA-based metrics have proved to be effective than other factual consistency evaluation metrics [12] in general summarization tasks and align more closely with human annotations [20, 39]. In this study, we adopt a QA-based approach as a backbone of our evaluation metric to verify factual consistency, incorporating answer extraction, question generation, question filtering, question answering, and answer overlap calculation modules.

Gold Answer Extraction (AE). The first step in QA-based factual consistency evaluation is to extract answer entities (keyphrases) from plain language summaries as gold answer, then verify factuality by comparing them with answers generated by a QA model for the same questions. If the generated answer is correct or relevant, the summary is considered factual. The previous study used PromptRank [40], a keyphrase generation method based on the `T5` model. To compare answer extraction strategies, we use PromptRank as the baseline, and evaluate LLM-based extractors, including

[3] `https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`
[4] `https://huggingface.co/datasets/MedRAG/statpearls`
[5] `https://huggingface.co/datasets/MedRAG/textbooks`

an open-source `Llama 3.1 8B Instruct` model (`Llama 3.1`) [41] and close-source `GPT-4o` (`GPT-4o`) [34]. Prompts for `Llama 3.1` and `GPT-4o` are in Appendix E.4.

Question Generation (QG). Given an input plain language summary, the QG model generates questions based on the extracted answers (§3.2.3) and the input plain language summary. Following previous studies [20, 21][6], we fine-tune `BART-large` model on standard QG datasets, including SQuAD [42] and QA2D [43], for use as the QG model in our evaluation metric. The QG model will generate multiple relevant questions based on the input, which will increase the probability of verifying the factual consistency by answering these questions in the following stages.

Question Filtering (QF). Questions generated by the QG model (§3.2.3) may not always be answerable. For example, the QG model generates "<What can occur over several days?>," which is vague and hard to predict the correct answer. To prevent these unanswerable questions from impacting the final evaluation performance, we follow QAFactEval [39] and remove the unanswerable questions using a pre-trained `Electra-large` model [44]. During QF, the filtering model receives only the plain language summary and its corresponding questions. In the subsequent QA stage (§3.2.3), answers are extracted from the source for answer overlap evaluation, while QF only determines whether the questions can be answered by the plain language summary.

Question Answering. The QA model extracts answers to answer the filtered questions from the source document. To prevent hallucinated output, we use an extractive QA model, a pre-trained `Electra-large`, which was the best performing QA model in QAFactEval [39].

Answer Overlap Evaluation (AOE). We evaluate the alignment between the gold and generated answers from the QA model using the baseline Learned Evaluation metric for Reading Comprehension (LERC) score [45] and BERTScore [46]. The final step in PLAINQAFACT is weighting the factual consistency scores for source simplification and elaborative explanation sentences. Specifically, the simplification score s is computed using only abstracts as the source in QA, while the explanation score e is calculated by incorporating both abstracts and retrieved knowledge as source for QA-based evaluation. n is the number of sentences in a plain language

[6]These studies use allennlp, an open-source NLP research library built on PyTorch, which is no longer actively maintained.

summary, after we split the summary using Natural Language Toolkit (NLTK)[7] and and classify each sentence with our fine-tuned classifier.

$$\textsc{PlainQAFact} = \frac{s_{\text{Avg.}} \cdot n_s + e_{\text{Avg.}} \cdot n_e}{n_s + n_e}, \tag{1}$$

where n_s is the number of simplification sentences, and n_e is the number labeled as explanation. $s_{\text{Avg.}}$ and $e_{\text{Avg.}}$ denote the average PLAINQAFACT scores computed for instances classified as simplification and explanation sentences, respectively.

4. Experiments

We conduct main experiments on three publicly available datasets, including PLAINFACT, CELLS [28], and FareBio [47]. To verify the effectiveness of our proposed PLAINQAFACT on PLS tasks, we compare with five widely used factual consistency evaluation metrics and two LLM-based evaluators.

4.1. Datasets

As introduced in Sec 3.1, we create PLAINFACT with fine-grained sentence-level annotations regarding the factuality types. Additionally, to test the generalizability of our pre-trained classifier, we also collect another subset from the test set of CELLS dataset [28], gathering 200 summary-abstract pairs. Similar to PLAINFACT (§3.1.1), we also conduct sentence-level perturbation for each plain language summary in our collected CELLS dataset. Therefore, we have 200 factual and 200 non-factual summary-abstract pairs for both PLAINFACT and CELLS datasets. We also compare PLAINQAFACT on the FareBio dataset, which contains 175 plain language summaries generated by seven different LLMs, together with the source articles [47]. Specifically, FareBio includes 25 full scientific articles, each summarized by seven LLMs, resulting 175 plain language summaries. Because FareBio does not include non-factual simplification sentences, we treat all simplification sentences as "factual." For each original summary, we form new plain language summaries by aggregating all sentences annotated as factual and all sentences annotated as non factual separately. This produces 33 non factual and 174 factual summaries. We also identify explanation sentences that receive the "factual hallucination" label (i.e., they are not verifiable from the source but are factually correct based on external knowledge) and build 53 factual explanation only summaries from them. We use the same set of 33 non factual summaries for both the main experiment (Table 2) and the explanation only evaluation (Figure I.6).

[7] `https://www.nltk.org/`

4.2. Experiment Settings

The classifier used in PLAINQAFACT is fine-tuned on PLAINFACT with a batch size of 32 for 10 epochs. We apply early stopping during training based on validation loss. The temperature is set to 0 for `GPT-4o` and 0.01 for `Llama 3.1`. The `Llama 3.1` model used in AE also uses a temperature of 0.01. For the QG model, we fine-tune `BART-large` with a batch size of 16, a learning rate of 3e-5, and two training epochs. The maximum input length is set to 512 for all models in PLAINQAFACT. More details of prompts and model hyper-parameters for `GPT-4o` and `Llama 3.1` are provided in Appendix E.

4.3. Existing Factuality Metrics

To the best of our knowledge, no prior work has efficiently evaluated factuality metrics for detecting elaborative explanations in PLS tasks. Moreover, most factuality metrics are designed for general-domain applications, largely due to limited quality and annotation of existing PLS datasets. To address this gap, we incorporate the following metrics in our experiment, informed by prior work [9]: (1) **Dependency-Arc Entailment (DAE)** [48] is an entailment-based method that evaluates summary factuality by breaking it into smaller entailment tasks at the arc level. The model independently determines whether each arc's relationship is supported by the input. (2) **AlignScore** [19] is a model-based factuality metric using `RoBERTa` [49]. It extracts claims from the summary and calculates the alignment scores with all the context chunks from the input document. The final score is an average of all highest alignment probabilities. (3) **SummaC** [16] is an NLI-based inconsistency detection method designed for summarization tasks. It segments documents into sentences and aggregates scores between sentence pairs using a trained NLI model. (4) **QAFactEval** [39] is a QA-based factuality evaluation metric that assesses summary consistency by generating questions from the source document and comparing the model-generated answers with expected answers. (5) **QuestEval** [50] is a reference-less evaluation metric employs a `T5` model to generate questions from the source document and verifies whether the summary can correctly answer them.

5. Results and Analysis

We first conduct a pilot study on the FactPICO dataset [9] to investigate the performance of existing factual consistency evaluation metrics on PLS tasks. Our pilot study (Appendix F) reveals two major limitations. First, many LLM-generated plain language summaries in FactPICO contain incomplete or incoherent sentences, highlighting the need for a benchmark based on high-quality, human-authored plain

language summaries. Second, existing automatic factual consistency metrics are not sensitive to evaluate the factuality of "added information," underscoring the need for a more domain-related factual consistency evaluation metric, especially for those summaries with additional explanations that are not mentioned by the original source texts. To address these challenges, we introduce a new expert-annotated dataset, PLAINFACT, and propose PLAINQAFACT, a QA-based, retrieval-augmented metric for evaluating factual consistency in plain language summaries.

In this section, we first benchmark PLAINQAFACT against five widely used automatic factual consistency evaluation metrics on three PLS datasets (§5.1). We then evaluate its performance in an explanation-only setting using PLAINFACT (§5.2), followed by an ablation study to assess the contribution of each component (§5.3). Next, we conduct an error analysis to examine failure cases and highlight directions for further improvement (§5.4).

5.1. Main Results

We report rank correlation (Kendall's τ), linear correlation (Pearson), and discrimination (AUC-ROC [51]) for three datasets (PLAINFACT, CELLS [28], and FareBio [47]) following previous studies [16, 39, 49] in Table 2. Overall, `GPT-4o` performs the best on CELLS and PLAINFACT, but its performance drops on FareBio. DAE is consistently the weakest across all datasets and metrics, suggesting that a dependency-only metric is insufficient for capturing the factual consistency required in PLS tasks.

On CELLS, PLAINQAFACT outperforms all other automatic metrics on all three criteria. Specifically, PLAINQAFACT achieves the best Kendall's τ (62.1) and Pearson (75.2) among all the automatic factual consistency evaluation metrics. As a comparative effect size, the pairwise difference in AUC-ROC between PLAINQAFACT and QAFactEval is small ($\Delta\text{AUC} - \text{ROC} = 0.3$, where Δ is PLAINQAFACT minus QAFactEval). After Holm–Bonferroni correction [52] on AUC-ROC, PLAINQAFACT shows significantly higher AUC-ROC than `Llama 3.1`, QuestEval, SummaC, DAE, and PromptRank ($p < 0.01$), whereas the differences with QAFactEval, AlignScore, and `GPT-4o` are not statistically significant (adjusted $p \geq 0.91$). This shows that our approach, first detecting elaborative content and then verifying it with retrieval, performs better than most of the existing reference-based metrics, which ask all questions from the source only. Similarly, on PLAINFACT, PLAINQAFACT achieves the highest linear correlation with human ratings among automatic metrics (Pearson= 81.3, 99% CI [76.3, 85.4]), surpassing `GPT-4o`. It ties AUC-ROC (96.4) score with QAFactEval and has a competitive Kendall's τ (65.7). As a comparative effect size, the pairwise difference in AUC-ROC between PLAINQAFACT and QAFactEval

Datasets	Metrics	Kendall's τ	Pearson	AUC (99% CI)
CELLS	Llama 3.1	57.5	42.1	83.4* [78.4, 88.0]
	GPT-4o	70.0	75.4	99.3 [98.6, 99.8]
	QAFactEval	61.6	70.6	93.5 [90.2, 96.3]
	QuestEval	23.2	28.3	66.4* [59.1, 73.3]
	SummaC	24.8	29.8	67.6* [60.3, 74.3]
	AlignScore	56.0	66.2	89.6 [85.3, 93.3]
	DAE	14.4	14.4	55.0* [50.7, 59.6]
	PLAINQAFACT	62.1	75.2	93.8 [90.3, 96.7]
FareBio	Llama 3.1	40.5	38.9	83.0 [69.0, 94.0]
	GPT-4o	27.3	35.7	75.7 [59.7, 89.8]
	QAFactEval	32.8	43.9	81.6 [69.2, 91.7]
	QuestEval	37.8	45.2	86.4 [78.0, 93.3]
	SummaC	33.8	32.1	82.5 [69.7, 92.9]
	AlignScore	32.2	47.2	81.0 [65.9, 93.0]
	DAE	8.8	8.8	53.4 [47.6, 57.6]
	PLAINQAFACT	16.8	35.6	66.1 [48.8, 82.1]
PLAINFACT	Llama 3.1	60.7	52.6	85.3* [80.6, 89.7]
	GPT-4o	69.8	80.3	99.2 [98.4, 99.8]
	QAFactEval	65.8	79.3	96.4 [94.0, 98.4]
	QuestEval	28.4	34.7	70.1* [63.1, 76.5]
	SummaC	34.1	42.8	74.1* [67.6, 80.4]
	AlignScore	60.2	72.6	92.5 [93.8, 98.4]
	DAE	5.3	5.3	51.7* [47.5, 55.9]
	PLAINQAFACT	65.7	81.3	96.4 [93.6, 98.5]

Table 2: Evaluation results of automatic metrics on the CELLS [28], PLAINFACT, and FareBio [47] datasets. Results are evaluated using Kendall's τ, Pearson correlation, and AUC-ROC (AUC), with eight metric scores compared against human-labeled factual consistency. The standard deviations (std.) over five runs are: 0.2 (PLAINQAFACT), 0.5 (`Llama 3.1`), and 6.0 (`GPT-4o`) on PLAINFACT; 0.1 (PLAINQAFACT), 0.2 (`Llama 3.1`), and 6.3 (`GPT-4o`) on CELLS; and 0.1 (PLAINQAFACT), 2.9 (`Llama 3.1`), and 17.0 (`GPT-4o`) on FareBio. * indicates a statistically significant difference compared to PLAINQAFACT ($p < 0.01$). **Note the plain language summaries in FareBio are generated by seven LLMs, while CELLS and PLAINFACT are written by the authors of original articles.**

is negligible ($\Delta\text{AUC} - \text{ROC} = -0.1$). These results on both datasets show that PLAINQAFACT is reliable for evaluating the factual consistency of plain language summaries. See Appendix G for more details of statistical tests.

On FareBio, no single method is best across all criteria. QuestEval achieves the best Kendall's τ (37.8) and AUC-ROC (86.4), while AlignScore achieves the best

Pearson correlation (47.2). However, FareBio differs from the other two datasets in several important ways: (1) the original sentence labels in FareBio are incomplete. Any sentence that can be traced back to the source is labeled as factual, so the dataset does not include sentences labeled as non-factual simplifications. This design leads to unreliable final summary-level labels. For example, even if a summary is labeled as factual and all of its sentences are labeled as factual, some sentences may still contain factual errors; (2) the processed dataset is highly imbalanced (33 non-factual vs. 174 factual summaries); (3) FareBio reports a fair Cohen's Kappa of 0.48 computed from 34 plain language sentences without adjudication, which indicates notable annotation quality issues in their curated dataset; and (4) all plain language summaries are generated by LLMs, and the plain language sentences are not guaranteed to be correct, since the annotation only checks whether the sentence is a source simplification without post-processed non-factual perturbation. These factors likely increase the difficulty of evaluation on FareBio and affect the performance of factual consistency evaluation metrics, including PlainQAFact and `GPT-4o`.

We also provide a detailed error analysis of FareBio that explains the main limitations of both PlainQAFact and the dataset (§5.5). In addition, because the non-factual plain language summaries of PlainFact and CELLS are generated by `GPT-4o` (§3.1.1), they may contain detectable artifacts that are not representative of real PLS errors. To further test the robustness of our proposed metric, we conduct additional experiments using the heuristic entity swap approach introduced in Guo et al. [12] to create non-factual plain language summaries. Specifically, following the same experiment setting as APPLS [12], we replace entities in each plain language sentence using the KBIN method [53], which replaces specific entity spans (such as names of viruses or diseases) with related but different concepts derived from the Unified Medical Language System (UMLS)[8] while maximizing NLI contradiction and minimizing LM perplexity. This approach ensures that each new perturbed sentence is not only different from the original sentence, but also logically contradicts it while remaining grammatically fluent. Overall, on the CELLS dataset, for both the LLM-perturbed setting and the heuristic entity swap method, our metric achieves the best performance across all three evaluation criteria compared with other automatic metrics, showing the robustness of our proposed metric. Detailed experiment results are reported in Appendix H.

In summary, PlainQAFact is a more consistent and effective metric for factual consistency compared to existing automatic approaches. It clearly improves over prior QA-based metrics on CELLS and matches or exceeds them on PlainFact. Al-

[8] `https://www.nlm.nih.gov/research/umls/`

though `GPT-4o` performs better in most of the settings, our PLAINQAFACT is built entirely on the open-source LLM (i.e., `Llama 3.1`), ensuring transparency, reproducibility, and accessibility. Its two-step evaluation process, detecting explanations and verifying them with retrieval, helps align metric scores more closely with human judgments in PLS evaluation tasks.

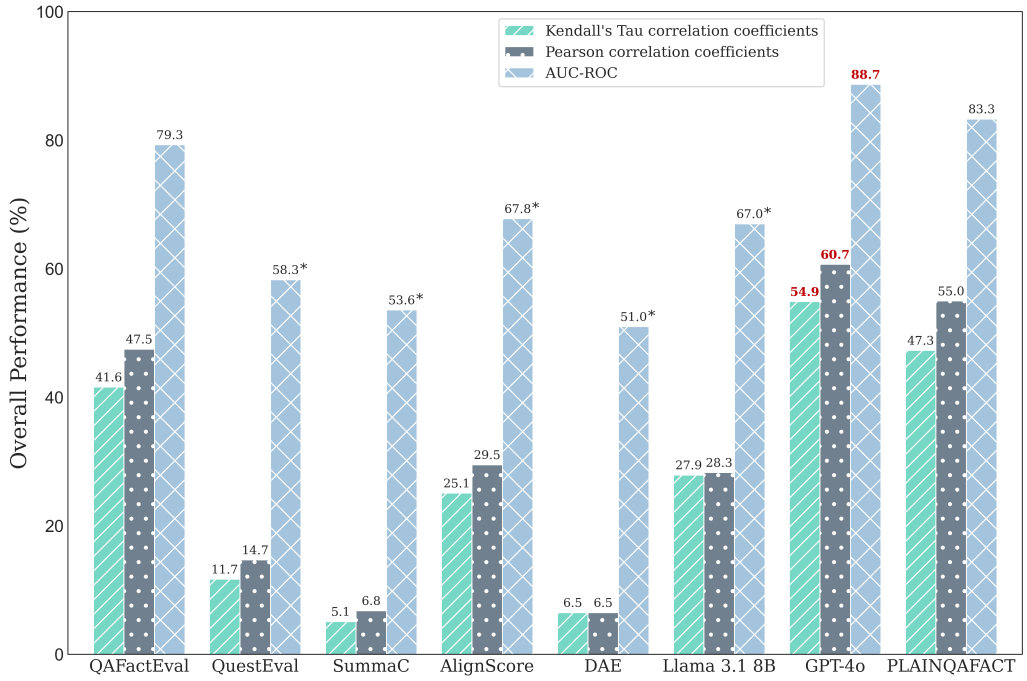


Figure 2: Overall performance on human-annotated elaborative explanation summaries from PLAINFACT (392 summaries). The std. of PLAINQAFACT, `Llama 3.1`, and `GPT-4o` are 0.1, 1.0, and 7.7, respectively based on five runs for each metric. * indicates a statistically significant difference compared to PLAINQAFACT ($p < 0.01$). PLAINQAFACT significantly outperforms most of the automatic factual consistency evaluation metrics in AUC-ROC. Note that the CELLS dataset does not contain annotations for elaborative explanations. Results of explanation-only evaluation on FactPICO and FareBio are reported in Appendix I.

5.2. Explanation-Only Evaluation

To test whether existing evaluation metrics are limited in assessing elaborative explanations in PLS, we evaluate PLAINQAFACT on human-annotated explanation-only summaries from PLAINFACT. We select only the sentences labeled as "elaborative explanation" and group them into plain language summaries, resulting in 392 summary–abstract pairs for PLAINFACT. Note that CELLS does not provide

sentence-level annotations for elaborative explanations, so we evaluate it only as a general PLS dataset (§5.1).

As shown in Figure 2, PLAINQAFACT achieves robust performance with an AUC-ROC of 83.3 (95% CI [77.7, 88.4]) and a Pearson correlation of 55.1, outperforming most automatic baseline metrics and approaching `GPT-4o` (AUC-ROC= 88.7, 99% CI [84.2, 92.8]). Comparison with open-source metrics, PLAINQAFACT demonstrates a notable improvement over standard open-source, QA- and NLI-based metrics. It produces statistically significant improvements ($p < 0.01$) over `Llama 3.1`, AlignScore, SummaC, QuestEval, and DAE across all correlation measures. For instance, PLAINQAFACT improves upon the zero-shot `Llama 3.1` baseline by a margin of ΔAUC-ROC= 16.3, showing that instruction-tuned LLM judges alone are not enough for reliable evaluation of elaborative content. Comparing with the state-of-the-art QA-based metric QAFactEval, PLAINQAFACT achieves a higher absolute AUC-ROC ($\Delta\text{AUC} - \text{ROC} = 4.0$) and Pearson correlation (Pearson= 55.1). However, after applying the Holm-Bonferroni correction, this improvement is on the margin of statistical significance ($p = 0.056$). These results indicate that PLAINQAFACT captures the factual consistency of elaborative explanations more effectively than most of the prior automatic metrics. While `GPT-4o` achieves higher overall scores in all three criteria, it shows much higher variance (std. 7.7) compared to PLAINQAFACT (std. 0.1), making it less stable for factual consistency evaluation.

Compared to the results in Table 2, AlignScore is competitive on general PLS tasks but drops on evaluating explanation-only sentences. In contrast, PLAINQAFACT remains consistently strong when we isolate explanation-only content. PLAINQAFACT detects added elaborative explanations and checks them with domain knowledge retrieval, which helps when factual consistency depends on facts beyond simple restatement from the source. Since 44% of sentences in our curated benchmark are labeled as elaborative by human annotators, we believe that PLAINQAFACT is the more suitable and robust factual consistency evaluation metric in this scenario.

5.3. Ablation Study

PLAINQAFACT consists of several modules, including a fine-tuned classifier, QA modules, a retrieval function, and an answer overlap evaluation process. To understand the contribution of each component, we conduct an ablation study on PLAINFACT. Table 3 summarizes the ablation results, where each component of PLAINQAFACT is individually modified to measure its impact on overall performance.

Fine-tuned Classifier Table 3 shows that removing the classifier and retrieving for every sentence **does not** improve factual consistency assessment. PLAINQAFACT

Component	Method Choice	τ	Pearson	AUC (99% CI)	std.
PLAINQAFACT		65.7	81.3	96.4 [93.6, 98.5]	0.2
Classifier	**Fine-tuned classifier**	-	-	-	-
	GPT-4o	62.8	77.1	94.3* [90.8, 97.1]	0.1
	No (retrieve for all)	61.2	74.2	93.2* [89.6, 96.2]	0.7
AE	**Llama 3.1**	-	-	-	-
	GPT-4o	66.9	81.9	97.2 [95.2, 98.9]	0.6
	PromptRank	66.3	80.2	96.0 [94.2, 98.4]	-
Retrieval Source	**ABS + TB + SP**	-	-	-	-
	TB + SP	63.4	76.8	94.8* [91.7, 97.3]	0.1
	ABS (no retrieval)	62.8	77.1	94.3* [90.8, 97.1]	0.1
Answer Overlap	**BERTScore**	-	-	-	-
	LERC	64.5	75.9	95.6 [92.7, 97.9]	0.1
Granular Level	**Sentence**	-	-	-	-
	Summary	49.3	59.6	84.7* [79.6, 89.5]	0.5

Table 3: Ablation study of PLAINQAFACT on PLAINFACT, analyzing the impact of individual components in PLAINQAFACT. τ: Kendall's τ; AE: answer extraction; ABS: abstract; TB: Textbooks; SP: StatPearls; AUC: AUC-ROC. The first row of each component setting represents our best combination. * indicates a statistically significant difference compared to PLAINQAFACT ($p < 0.01$). We run each setting for five times and report the standard deviation (std.). Note that only the settings using LLMs may produce fluctuated scores.

statistically significantly outperforms the settings using `GPT-4o` as the classifier and the setting without a classifier ($p < 0.01$). Performance drops across all criteria (Kendall's τ from 65.7 to 61.2; Pearson from 81.3 to 74.2; AUC-ROC from 96.4 to 93.2), and the std. increases to 0.7. This approach also requires notably more computation. Using `GPT-4o` as a simple classifier provides reasonable results but still underperforms compared to our fine-tuned classifier. These findings support the use of a lightweight fine-tuned classifier to trigger retrieval only when necessary.

Answer Extraction For extracting gold answers from plain language summaries, the LLM based extractors perform better than PromptRank [40]. `GPT-4o` achieves the best overall scores (Kendall's τ=66.9, Pearson=81.9, AUC-ROC=97.2), with only small gains over our Llama-based method. In PLAINFACT, PromptRank obtains a slightly lower Pearson correlation than PLAINQAFACT. However, in the ablation study on the CELLS dataset in Table H.10, PLAINQAFACT significantly outperforms PromptRank in both perturbation settings (see details in Appendix H). Overall, we do not find a statistically significant difference between PLAINQAFACT and the other answer extraction methods. PLAINQAFACT is statistically tied with

PromptRank (AUC-ROC=96.0, $p = 0.34$) and with `GPT-4o` AE (AUC-ROC=97.2, $p = 1.0$). Although `GPT-4o` achieves slightly higher performance overall, the marginal gain does not justify the increased cost, making `Llama 3.1` a more practical and cost-effective choice for large-scale factual consistency evaluation.

Component	Method Choice	τ	Pearson	AUC (99% CI)	std.
Simplification	ABS	63.5	79.0	94.8 [91.6, 97.4]	0.2
Explanation	Abs	32.8	38.2	72.7* [65.8, 79.2]	0.2
	TB	31.2	37.2	74.6* [68.0, 80.9]	0.1
	SP	33.3	37.8	72.5* [65.9, 79.1]	1.7
	TB + SP	36.7	42.0	75.4* [68.8, 81.7]	0.1
	ABS + TB	40.3	47.0	78.1* [71.8, 84.1]	0.2
	ABS + SP	40.5	47.4	78.3* [71.7, 84.2]	0.2
	ABS + TB + SP	**44.2**	**51.3**	**81.0** [74.9, 86.7]	0.1
Full Dataset	ABS	62.8	77.1	94.3* [90.8, 97.1]	0.1
	TB	60.8	73.8	94.0* [90.7, 96.7]	0.1
	SP	61.3	74.2	93.3* [89.6, 96.4]	0.1
	TB + SP	63.4	76.8	94.8* [91.7, 97.3]	0.1
	ABS + TB	64.3	79.7	95.4* [92.2, 97.9]	0.2
	ABS + SP	65.0	80.3	95.9 [92.9, 98.2]	0.2
	ABS + TB + SP	**65.7**	**81.3**	**96.4** [93.6, 98.5]	0.2

Table 4: Ablation study results on the PLAINFACT, evaluating how different retrieval sources affect various information types using PLAINQAFACT. The fine-tuned classifier categorizes input sentences as **source simplification** or **elaborative explanation**. Overall, the numbers of factual summaries that only include source simplification and elaborative explanation sentences are 198 and 191 respectively, and 198 and 194 for non-factual summaries. Simplification: Source Simplification; Explanation: Elaborative Explanation; τ: Kendall's τ; ABS: abstracts; TB: Textbooks; SP: StatPearls; AUC: AUC-ROC. * indicates a statistically significant difference compared to ABS+TB+SP ($p < 0.01$). We run each setting for five times and report the standard deviation (std.) in the brackets.

Combined Domain Resources for Retrieval As shown in Table 3, combining all three evidence sources (ABS+TB+SP) significantly outperforms the other two settings ($p < 0.01$). This suggests that comprehensive retrieval, which includes the original source abstracts, improves overall factual consistency assessment. Table 4 shows the retrieval ablation study results in detail. In the full dataset setting, ABS+TB+SP achieves AUC-ROC=96.4 and Pearson=81.3, outperforming all retrieval variants that use only a subset of the retrieval corpus, including TB+SP (AUC-ROC=94.8), TB only (94.0), SP only (93.3), and ABS+TB (95.4). To further examine the effectiveness of retrieval for source simplification and elaborative explanation, Table 4 breaks down the results as follows:

(1) *Source simplification*: These are sentences in plain language summaries classified as "source simplification." We evaluate summaries containing only simplification sentences, using abstracts as the source. Using abstracts as sole source texts achieves strong performance (63.5/79.0/94.8) across all three criteria.

(2) *Elaborative explanation*: For summaries containing only explanation sentences, abstracts alone are not sufficient, performing much worse than in simplification cases. Adding abstracts with external sources improves performance, with the best results obtained by combining all three sources. ABS+TB+SP outperforms the abstract-only baseline ABS (AUC-ROC=72.7) and all partial-view variants (TB+SP, TB, SP, ABS+TB, and ABS+SP). The gain over ABS is substantial, with $\Delta\text{AUC} - \text{ROC} = 8.3$ on (99% CI [3.7, 13.0]), and similarly large improvements in correlation.

(3)*Full dataset*: Retrieval from StatPearls provides better results than retrieval from Textbooks, highlighting the importance of using high-quality, domain-specific knowledge bases for PLS evaluation. Overall, the best combination across all settings is **ABS+TB+SP** (65.7/81.3/96.4). Relative to the abstract-only baseline, ABS (AUC-ROC=94.3), the gain in discrimination is $\Delta\text{AUC} - \text{ROC} = 2.1$ (99% CI [0.5, 4.2], where Δ is ABS+TB+SP minus ABS).

These findings show that effective retrieval for factual consistency should include both the source abstract and reliable external medical references, especially when summaries contain elaborative explanations.

Answer Overlap Evaluation BERTScore ties with the LERC approach introduced in QAFactEval [21], showing no significant difference in AUC-ROC ($p = 0.29$). BERTScore achieves slightly higher nominal values in both correlation metrics. There are fundamental differences between the two metrics. LERC is a learned model trained to predict human factual consistency scores, which may reduce its generalizability to new domains or sentence types. In contrast, BERTScore computes embedding-based semantic similarity, allowing it to capture fine-grained semantic overlap between generated and gold answers. Since PLAINQAFACT evaluates factual consistency by comparing model-generated answers to those extracted from the source, BERTScore's sensitivity to semantic alignment makes it a more effective and robust choice for this step of the evaluation.

Sentence-level vs. Summary-level Evaluation To further examine how input granularity affects factual consistency evaluation, we deactivate the sentence-splitting function and instead pair each plain language summary with its abstract to evaluate factual consistency at the summary level. Our best sentence level evaluation achieves an AUC-ROC of 96.4, which is significantly higher than the summary level setting ($p < 0.01$). The summary level setting performs worse across all three criteria

(Kendall's τ=49.3, Pearson=59.6, AUC-ROC=84.7; std. 0.5). Evaluating at the summary level introduces broader contextual dependencies, which can omit sentence-specific factual errors or create inaccurate entailments. These results suggest that processing one sentence at a time makes QA-based factual consistency evaluation more focused and reduces noise from unrelated context.

5.4. Error Analysis on PLAINFACT

To analyze cases where PLAINQAFACT fails on the PLAINFACT benchmark, we categorize one correct example and three types of errors in Table 5. For each case, we present the original plain language sentence, the corresponding scientific abstract with retrieved knowledge (from medical Textbooks and StatPearls), the model-generated questions based on extracted answers, and the QA model's final responses. Each question is generated based on the extracted answer and its corresponding plain language sentence.

In the correct case, the QG model generates a question from the plain language sentence and the extracted answer "SSB." The QA model then provides the correct answer using the retrieved knowledge rather than the original abstract, demonstrating PLAINQAFACT 's effectiveness in evaluating explanation sentences.

In the first failure case, the extracted answer is "pregnancy," but the QA model returns "cycle control" from the abstract because the correct term is missing in the retrieved content. Since this sentence is classified as an "elaborative explanation," which requires external knowledge for verification, this error points to a potential misclassification between *simplification* and *explanation* by the fine-tuned classifier. The second failure illustrates how noisy retrieved knowledge can impair evaluation. The QA model provides an irrelevant answer from the retrieved content that does not match the extracted gold answer. The third case shows an unanswerable question generated by the QG model (e.g., regarding "Government"), highlighting a limitation of QA-based factual consistency evaluation. Some questions remain unanswered even with retrieval. Additionally, we acknowledge that some plain language sentences may not generate any questions, resulting in empty question sets. Addressing these challenges requires further improvements in classification, retrieval, and domain-specific question generation.

5.5. Error Analysis on FareBio

As shown in Table 6, we provide several failure cases on the FareBio dataset based on the results in Table 2. We rank the PLAINQAFACT scores for all summaries and select two examples with the lowest scores among those labeled NF and two examples with the lowest scores among those labeled F. The first NF example in Table 6 shows

Correct Case	**Plain Language Sentence:** Limits to the availability of SSB in schools (e.g. replacing SSBs with water in school cafeterias). [54] **Scientific Abstract:** Frequent consumption of excess amounts of sugar-sweetened beverages (SSB) is a risk factor for obesity, type 2 diabetes, cardiovascular disease and dental caries... [54] **Generated Question:** Limits to the availability of what in schools? **Retrieved Knowledge:** Even though additional data is required to determine the impact of limiting the availability of nutrient-poor or high-sugar goods in schools on obesity, some study results have shown a net-positive result... **Final Answer:** nutrient-poor or high-sugar goods
Misclassification	**Plain Language Sentence:** This review looked at how well the methods worked to prevent pregnancy, if they caused bleeding problems, if women used them as prescribed, and how safe they were. [55] **Scientific Abstract:** To compare the contraceptive effectiveness, cycle control, compliance (adherence), and safety of the contraceptive patch or the vaginal ring versus combination oral contraceptives (COCs)... [55] **Generated Question:** This review looked at how well the methods worked to prevent what? **Retrieved Knowledge:** Appropriate treatment for the underlying etiology should start as soon as possible, and the patients and family members should receive appropriately targeted education... **Final Answer:** cycle control
Retrieval Quality	**Plain Language Sentence:** The patch is a small, thin, adhesive square that is applied to the skin. [55] **Scientific Abstract:** Users of the norelgestromin-containing patch reported more breast discomfort, dysmenorrhea, nausea, and vomiting. In the levonorgestrel-containing patch trial, patch users reported less vomiting, headaches, and fatigue...[55] **Generated Question:** The patch is a small, thin, what kind of square applied to the skin? **Retrieved Knowledge:** Nonstick dressing Petrolatum-infused gauze strip or other material to form a bolster over the graft site. This may be sutured or taped securely in place to provide some pressure and to keep graft immobilized. **Final Answer:** Nonstick dressing Petrolatum-infused gauze strip
Unanswerable Question	**Plain Language Sentence:** Government officials, business people and health professionals implementing such measures should work together with researchers to find out more about their effects in the short and long term. [54] **Scientific Abstract:** To assess the effects of environmental interventions (excluding taxation) on the consumption of sugar-sweetened beverages and sugar-sweetened milk, diet-related anthropometric measures and health outcomes, and on any reported unintended consequences or adverse outcomes...[54] **Generated Question:** What officials, business people and health professionals implementing such measures should work together with researchers to find out more about their effects in the short and long term? **Retrieved Knowledge:** Implementation should be accompanied by high-quality evaluations using appropriate study designs, with a particular focus on the long-ter effects of approaches suitable for large-scale implementation. **Final Answer:** long-ter effects

Table 5: Error analysis of PLAINQAFACT with intermediate metric outputs. "Retrieved Knowledge" refers to the source texts retrieved for each plain language sentence. We present correct and failure cases sampled only from elaborative explanation examples (i.e., sentences classified as explanations during evaluation). The correct case has a PLAINQAFACT score above 0.6, while the failure cases have scores below 0.5.
Color legend: extracted answer, answer origin, correct answer, incorrect answer.

a true factual error. The target plain language sentence changes the numerical range and introduces the value zero, which does not appear in the source. However, our answer extractor does not reliably extract the exact numbers, so PLAINQAFACT does not capture the fact errors. The second NF example shows a label noise in FareBio.

Type	Target PLS Sentence	QA Extraction Analysis	Rationale
NF	This performance was identical in the comparative analysis subgroup and was within the range of physicians at different levels of expertise: 0.86-0.97 and 0.	**Ref:** physicians **Q:** *This performance was identical in the comparative analysis subgroup and was within the range of what at different levels of expertise?* **Pred:** physicians	The non-factual error is a value mistake. The target sentence incorrectly truncates the specificity range to "0," which does not appear in the source and therefore being labeled as non-factual. However, our answer extractor fails to identify the numerical values as gold answer to verify its factual consistency.
NF	Imaging also showed chronic compression of the coeliac axis with compensatory hypertrophy of the gastroduodenal artery.	**Ref:** gastroduodenal artery **Q:** *Imaging also showed chronic compression of the coeliac axis with compensatory hypertrophy of what artery?* **Pred:** gastroduodenal	The target sentence correctly restates information that is directly present in the source, but it is labeled as non-factual in FareBio.
F	In summary, the article discusses the potential impact of COVID-19 on the musculoskeletal system... and emphasizes the need for further research to understand the underlying mechanisms and develop appropriate interventions.	**Ref:** interventions **Q:** *In summary, the article discusses the potential impact of COVID-19 on the musculoskeletal system, particularly in the context of aging and Long-COVID, and emphasizes the need for further research to understand the underlying mechanisms and develop appropriate what?* **Pred:** multidisciplinary approach	The predicted answer comes from the original source. This target PLS sentence is classified as a simplification; however, the term "interventions" does not appear in the source.
F	Some patients experience persistent symptoms after recovering from the initial infection, a condition known as "Long-COVID."	**Ref:** infection **Q:** *Some patients experience persistent symptoms after recovering from what condition known as Long-COVID?* **Pred:** SARS-CoV-2	In the source, SARS CoV 2 always appears together with the word "infection." Our QA model captures the relevant answer, but our answer overlap evaluation approach does not align this prediction with the gold answer, which results in a lower PlainQAFact.

Table 6: Error analysis of PlainQAFact on the FareBio dataset. NF/F = non-factual or factual; Ref = gold/extracted answer; Q = generated question; Pred = predicted answer.

The target sentence restates information that is clearly present in the source, but the sentence is labeled as NF. Third, we find pipeline errors even when the FareBio label is factual. In the third example, the target sentence correctly summarizes the current knowledge and is classified as simplification by our classifier, but the answer extractor extracts the word "interventions," which does not appear in the source and thus leads to a low score. In the fourth example, the QA module predicts "SARS CoV 2" as the answer, which is consistent with the "infection" described in the plain

language sentence, but the final answer overlap evaluation step cannot match this prediction to the gold answer "infection." Overall, we conclude that most errors of our PLAINQAFACT come from the answer extraction and answer overlap evaluation stages in the FareBio dataset, while the dataset contains noisy factuality labels.

To mitigate these errors (§5.4, §5.5), we provide several directions for future work. First, misclassification of plain language sentences by the factuality type classifier can be reduced by training on more annotated examples or use advanced low-resource model tuning strategies for text classification [56]. Second, retrieval errors can be mitigated by conducting precise filtering method to select the most relevant snippets, such as limiting the length of retrieved snippets and chunk size, using a combination of dense and sparse retrieval, and expanding the domain corpus (e.g., PubMed) to cover more specialized topics. Third, to handle unanswerable or vague questions, future work can improve the question filtering stage that removes questions that do not have clear grounding in both source retrieved context, or that are too general to support reliable factual checks using heuristic approaches. Finally, for the answer overlap evaluation module, more advanced entailment-based models can be applied to replace the BERTScore semantic matching to check whether the predicted answer is consistent with the gold answer after simple normalization of entities and synonyms. We report the quantitative error analysis of FareBio in Appendix J.

6. Discussion and Conclusion

Our study advances the assessment of hallucinations in the field of PLS by supplying both the first domain expert-annotated biomedical PLS benchmark PLAINFACT and a novel retrieval-augmented QA-based factual consistency evaluation metric PLAINQAFACT. Unlike existing biomedical domain corpora that either lack fine-grained labels for elaborative content [57] or focus solely on text simplification (e.g., FactPICO[9]), PLAINFACT captures sentence-level distinctions, including simplification versus explanation, functional roles, and explicit alignment to source sentences. This granularity not only enables precise error analysis of hallucinations introduced for clarity, but also provides a reusable framework for researchers in other disciplines (e.g., legal, technical) to replicate our annotation protocol and build high-quality factual consistency datasets tailored to their specialized texts. By offering these data and detailed annotation protocol (Appendix A), we anticipate community-driven extensions, such as multilingual adaptations or integration with domain-specific ontologies that will democratize development of trustworthy summarization models.

Building on this resource, PLAINQAFACT differentiates from prior QA-based metrics (e.g., QAFactEval[21], QuestEval[50]) by first classifying whether a sentence

requires external verification given elaborative explanations are often ignored by metrics relying solely on source text, and improve the evaluation accuracy without retrieving for every plain language summary instance. Then, retrieving targeted domain knowledge for each elaborative explanation before posing and answering fact-checking questions. The ablations demonstrate that selective retrieval yields substantial gains in Kendall's τ, Pearson correlation, and AUC-ROC over both end-to-end LLM judges[41] and alignment-based scorers[19], highlighting that judicious incorporation of external evidence is essential for robust factual consistency assessment. Beyond its empirical strengths, PLAINQAFACT serves as a plug-and-play evaluation metric for future summarization systems. Developers can leverage our classifier to classify sentences, apply retrieval only where necessary, thus containing computational costs and obtain interpretable question–answer pairs that manifest non-factual hallucinations. More importantly, PLAINQAFACT is developed with the open-source backbone model (i.e., `Llama 3.1`), ensuring broad use with transparency, reproducibility, and lower computational costs. Looking forward, extending our proposed evaluation method to interactive human–AI workflows, where clinicians or subject-matter experts verify QA outputs on-the-fly, could further enhance trust and adoption in high-stakes domains, including medicine, healthcare, or law.

Despite these promising results, we consider the following limitations of this study. (1) Our error analysis reveals that misclassifications, particularly between source simplifications and elaborative explanations, can lead to retrieval mismatches and unanswerable QA prompts. These issues underscore the need for more nuanced classification and question generation modules. Although we provide substitute solutions for classifying input summaries or sentences, our fine-tuned classifier is limited to the biomedical domain and specifically designed for PLS tasks, fine-tuned through PLAINFACT. We acknowledge that the dataset used for model fine-tuning is insufficient, resulting in only moderate classification accuracy. Given the scarcity of high-quality human-annotated data for classifying source simplification and elaborative explanation information, future efforts are needed on developing domain-specific (e.g., law, healthcare, social science, etc.) classifiers [56] for factual consistency evaluation. (2) While our experiments indicate that our retrieval-augmented evaluation metric can improve factual consistency assessment in most of the settings, especially in elaborative explanation evaluation, the computational time increases compared to NLI-based evaluation metrics and LLM-based evaluators. The trade-off between evaluation precision and efficiency suggests that further optimization will be beneficial. We suggest to balance wisely based on the trade-offs we report in this study regarding the evaluation time and accuracy. PLAINQAFACT is a research-oriented metric that prioritizes interpretability and transparency over speed. (3) Given the

fair IRA scores, especially for the factuality type field, we acknowledge that our classification method is not fully reliable. The annotated dataset reflects subjective judgments, particularly when distinguishing source simplification from elaborative explanation, and the training set is limited in size. We advise users to apply our pre-trained classifier with caution when using it for later factual consistency evaluation steps.

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Appendix A. Dataset Annotation Protocol

We developed a comprehensive annotation procedure for freelancers on Upwork to conduct fact-checking annotations.

Our annotation procedure involves two stages, starting with thorough training using detailed examples to ensure consistent understanding of the task between annotators. Annotators receive a spreadsheet where each row contains a pair of data: a sentence extracted from the plain language summary and its corresponding scientific abstract. For every sentence-abstract pair, the annotators are required to label three features: External, Category, and Relation, with the appropriate labels.

Step 1: Sentence Annotation Instructions

Compared to the scientific abstract, analyze each sentence of the plain language summary across three dimensions: external information, category, and relation.

1. **External**: Determine whether the sentence includes information does not present in the scientific abstract.

   1) `Yes`: The sentence contains external information that is not explicitly mentioned, paraphrased, or implied in the scientific abstract.

   2) `No`: The sentence contains information that is explicitly stated or closely paraphrased from the scientific abstract.

2. **Category**: Classify the sentence of the plain language summary into one of the following categories (you can only choose one category per annotation)

   1) `Definition`: Provides a fundamental explanation of a term.

   2) `Background`: Information that helps understand the term within the context of the abstract, such as relevance, significance, or motivation.

   3) `Example`: Specific instances that illustrate the use of the term in the scientific abstract.

   4) `Method/result`: Details about the methodology or results described in the scientific abstract.

   5) `Other`: For sentences that do not fit into the categories above, please indicate the category

3. **Relation**: Identify the sentence(s) in the scientific abstract that the sentence of the plain language summary is related to. Use indices of sentences from the scientific abstract to link the sentence of the plain language summary. You can select one or more sentences from the scientific abstract. List the indices like s1_1,s2_3,s3_6. If no relation is found, mark it as "external."

In Step 2, we provide explanations of the existing sentence-level indexes for plain language sentences and scientific abstracts.

Step 2: Using the Annotation Spreadsheet

You will work within a structured spreadsheet containing the segmented sentences from both the summaries and the scientific abstracts.

1. **Target_Summary_ID**: A unique identifier for each plain language summary. Sentences from the same plain language summary share the same ID.

2. **Target_Sentence_Index**: An identifier for each sentence within a summary, forming as tx_y, where 'tx' is the same as its 'Target_Summary_ID', and 'y' represents the index of each sentence in the plain language summary, starting from 1. e.g., t0_1 refers to the first sentence of the first summary)

3. **Target_Sentence**: The plain language sentence you are annotating.

4. **Original_Abstract**: The abstract corresponding to each summary, with each sentence indexed for easy reference.

Each annotator will annotate 40 summary-abstract pairs to ensure each sentence of the plain language summary has two sets of annotations from different people. For each row of the spreadsheet, they need to annotate three columns: "External," "Category," and "Relation."

Annotation Fields to Be Completed

<TO BE ANNOTATED> **External**: Mark "Yes" or "No" to indicate if the sentence in the 'Target_Sentence' column contains external information.

<TO BE ANNOTATED> **Category**: Choose the most fitting category of the sentence from the list (Definition, Background, Example, Method/Result, Other).

<TO BE ANNOTATED> **Relation**: List the relevant sentence indices from the abstract in the 'Original_Abstract' column that relate to the plain language sentence (e.g., s10_1,s10_5). For example, filling in 's10_1,s10_5,s10_9' if you think these three sentences from the abstract are relevant to sentence t1_3. Use commas to separate multiple indices.

To ensure annotators fully understand the context and task requirements, we provide comprehensive annotation training and a screening test prior to the annotation process. We select candidates from freelancers with a medical education background,

and only those who pass the screening test are finalized as annotators.

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Appendix B. Dataset Examples

In Table B.7, we presents three representative examples from PLAINFACT. Each example is annotated with five features: a plain language sentence, an indicator of whether the sentence is simplification or explanation, its category, its relation, and the corresponding abstract. All plain language sentences are **factual**, as they were written by the authors from the Cochrane database. "Need External Information?" feature specifies whether a sentence can be validated solely by the abstract. A "Yes" label indicates that the sentence includes information not explicitly mentioned in the abstract, and vice versa. The "Relation" feature identifies the sentence(s) in the scientific abstract most relevant to the plain language summaries; if no corresponding content exists in the abstract, the relation is marked as "external." We randomly sample three examples from the dataset to to illustrate the dataset's structure. Additionally, indexes have been created for both the plain language sentences and the abstract sentences to facilitate annotation.

Appendix C. Qualitative Analysis of PLAINFACT

Given the fair inter-rater agreement for the "Need External Information?" field in PLAINFACT, we conduct a targeted qualitative error analysis. We randomly sample one representative plain language sentence from each of five annotation batches where the two annotators disagree. Two annotators independently judge whether external information is required, and a third judge provides a brief rationale. As shown in Table C.8, most disagreements fall into two categories: (i) differing interpretations of whether an elaborative explanation introduces new information beyond the source, and (ii) failures to locate or recognize supporting evidence in the source abstract. These examples highlight that the "Need External Information?" decision often depends on subtle judgments about sufficiency and specificity of source coverage, suggesting that clearer guidelines and additional training examples may further improve annotation consistency.

	Feature	Annotation
Example 1	Plain Language Sentence	Gout caused by crystal formation in the joints due to high uric acid levels in the blood.
	Need External Information?	yes
	Category	Background
	Relation	external
	Corresponding Abstract	None
Example 2	Plain Language Sentence	Reducing blood pressure with drugs has been a strategy used in patients suffering from an acute event in the heart or in the brain, such as heart attack or stroke.
	Need External Information?	yes
	Category	Background
	Relation	s10_1,s10_2
	Corresponding Abstract	<s10_1>Acute cardiovascular events represent a therapeutic challenge. <s10_2>Blood pressure lowering drugs are commonly used and recommended in the early phase of these settings.
Example 3	Plain Language Sentence	We looked at whether choice of antibiotic made a difference in the number of people who experienced failed treatment, and we determined the proportions who had resolution of fever at 48 hours.
	Need External Information?	no
	Category	Method/Result
	Relation	s15_16,s15_17,s15_20
	Corresponding Abstract	<s15_16>For treatment failure, the difference between doxycycline and tetracycline is uncertain (very low-certainty evidence). <s15_17>Doxycycline compared to tetracycline may make little or no difference in resolution of fever within 48 hours (risk ratio (RR) 1.14, 95% confidence interval (CI) 0.90 to 1.44, 55 participants; one trial; low-certainty evidence) and in time to defervescence (116 participants; one trial; low-certainty evidence). <s15_20>For most outcomes, including treatment failure, resolution of fever within 48 hours, time to defervescence, and serious adverse events, we are uncertain whether study results show a difference between doxycycline and macrolides (very low-certainty evidence).

Table B.7: Examples of our curated dataset. Need External Information feature represents whether a plain language sentence is a simplification or an explanation. A label of "yes" indicates that the sentence is an explanation and requires additional elaborative information beyond the source abstract to verify its factual consistency. Conversely, a label of "no" shows that the sentence is a simplification that can be validated using only the source abstract.

Appendix D. Data and Indexing Statement

In this study, we follow the same protocol as used in MedRAG [37] to conduct sentence-level retrieval of the PLS sentences. Specifically, we use top-K as 3 across

ID	Target PLS Sentence	A1	A2	Final	Rationale
Ex1	Obesity is associated with many health problems and a higher risk of death.	✗	✓	✓	"Health problems" and "higher risk of death" are not in the original abstract. No association between obesity and health/death issues exists in the source.
Ex2	It can be hard for the person to gain control of the eye, which can cause distress in social situations.	✓	✗	✓	Most relevant source (from A1): *"Spontaneously manifest DVD can be difficult to control and often causes psychosocial concerns."* However, no direct correlation between eye control and DVD exists; social situations are not equal to psychosocial concerns.
Ex3	Most participants were middle-aged.	✓	✗	✗	A2 provides supporting source: *"Applicability of evidence was limited to middle-aged participants who were relatively free of co-morbidity and were undergoing elective abdominal surgery."* This sentence is supported by the source.
Ex4	In some people, winter blues becomes depression, which seriously affects their daily lives.	✓	✗	✓	A2 highlights relevant source: *"SAD is a seasonal pattern of recurrent major depressive episodes that most commonly occurs during autumn or winter and remits in spring."* However, no causal relation between winter blues and depression in source.
Ex5	RDS is the most common cause of disease and death in babies born before 34 weeks' gestation.	✗	✓	✓	Most relevant source: *"RDS is the single most important cause of morbidity and mortality in preterm infants."* However, "babies born before 34 weeks' gestation" is not mentioned in the abstract.

Table C.8: Qualitative analysis of PLAINFACT annotation examples. $A1/A2$ = Annotator 1 or 2; ✓/✗ indicate yes or no judgments for the "Need External Information?" field.

all of our experiments on both StatPearls[9] and medical Textbooks[10] [38] provided in the MedRAG repository[11]. Specifically, the medical Textbooks corpus comprises 18 widely used medical textbooks (e.g., for USMLE preparation) originally collected by Jin et al. [38], while StatPearls corpus consists of point-of-care clinical decision support articles. The authors of MedRAG collect 9,330 publicly available articles from the NCBI Bookshelf[12], which permits redistribution for non-commercial research purposes. The StatPearls corpus applies a hierarchical approach for chunking, treating each paragraph in an article as a snippet and splice all relevant hierarchical headings to form the corresponding title. The medical Textbooks is processed into chunks of no more than 1,000 characters. These corpus are made publicly available for research replication via the MedRAG HuggingFace repository[13] under the original authors'

[9] `https://huggingface.co/datasets/MedRAG/statpearls`
[10] `https://huggingface.co/datasets/MedRAG/textbooks`
[11] `https://github.com/Teddy-XiongGZ/MedRAG?tab=readme-ov-file#corpus`
[12] `https://www.ncbi.nlm.nih.gov/books/NBK430685/`
[13] `https://huggingface.co/MedRAG`

distribution terms. We do not redistribute raw StatPearls and Textbooks content beyond the MedRAG release and rely exclusively on their officially published corpus to ensure replicability. Researchers wishing to replicate the corpus index should (i) access the same StatPearls[14] and Textbooks[15] datasets, (ii) ensure their use complies with the copyright and permissions of the source license of MedRAG [37] and MedQA [38]. Following MedRAG, we use the same `RecursiveCharacterTextSplitter` from the LangChain[16] library to perform chunking. We directly retrieve top 3 most relevant snippets from both corpus. For the retriever, we use MedCPT [36] (specifically the `ncbi/MedCPT-Query-Encoder`[17] version, commit hash: `d83a36c`) for dense retrieval with CLS token pooling. Documents are indexed using FAISS [58] with an `IndexFlatIP` configuration (exact inner product search, , no quantization or clustering) for 768-dimensional embeddings. The retrieval system computes the inner product between query and document vectors to identify the most relevant snippets. This matches the default MedRAG implementation for MedCPT.

Appendix E. Reproducibility

Appendix E.1. Fine-tuning QG Model

We fine-tune the QG model (i.e., `BART-large`) with the same datasets used in QAFactEval [39], including SQuAD [42] and QA2D [43]. We use a batch size of 16, a learning rate of 3e-5, maximum sequence length of 512 tokens, and two training epochs for QG model fine-tuning.

Appendix E.2. Fine-tuning Factuality Type Classifier

As mentioned in Section 3.2.1, we fine-tune a classifier using `PubMedBERT-base` model through our curated PlainFact dataset. Specifically, we set the random seed to 42 to split PlainFact into training, validation, and test sets. We tune the `PubMedBERT-base` model for three epochs with early stopping under the validation set. The final fine-tuning accuracy of the classifier on the test set is 0.77. The same seed (42) is also used to sample 200 summary-abstract pairs from the CELLS test set for the comparison in Table 2. The classifier used in PlainQAFact is fine-tuned on PlainFact with a batch size of 32 for 10 epochs. We apply early stopping during training based on validation loss.

[14] `https://huggingface.co/datasets/MedRAG/statpearls`
[15] `https://huggingface.co/datasets/MedRAG/textbooks`
[16] `https://www.langchain.com/`
[17] `https://huggingface.co/ncbi/MedCPT-Query-Encoder`

Appendix E.3. General Details

For other open-source models used in this study, we apply the default version. For the QA model, we apply the original `Electra-large` model[18] used in QAFactEval [39]. All experiments we conduct are under one NVIDIA A100 GPU with 40 GB GPU memory. We employ the Natural Language Toolkit (NLTK)[19] to split the plain language summaries into sentences using the `punkt` package. For question filtering strategy, we follow the same protocol in QAFactEval. The temperature is set to 0 for GPT-4o and 0.01 for Llama 3.1. The Llama 3.1 model used in AE also uses a temperature of 0.01. The maximum input length is set to 512 for all models in PlainQAFact.The QA model (i.e., `Electra-large`) used in our PLAINQAFACT is directly downloaded from QAFactEval [39]. However, we set its maximum input length to 512 (from 364) tokens to incorporate more source context. Similarly, to ensure that the answers extracted during the AE stage are valid (i.e., within the 512-token limit to maintain consistency with the subsequent QA model), we configure the LLMs' input lengths to 512 tokens. In MedCPT retrieval, we set `k` (the number of retrieved snippets) to 3, ensuring the retrieved information remains within a short context window for the subsequent QA process (Appendix E). In the QF stage, we follow QAFactEval [39] and remove the unanswerable questions using the pre-trained `Electra-large` QA model[20].

We apply the default setting (i.e., `RoBERTa-base`) for AlignScore [19] and SummaC-Conv for SummaC [16]. For `GPT-4o` experiments as a factual consistency evaluator, we set the temperature as 1.0 in this paper. For the FareBio dataset, each plain language sentence is annotated with two binary labels: faithfulness and factual hallucination. In the released data, there are no sentences that are faithful but factually wrong. For the dataset pre-processing, we treat a sentence as not factual only when both labels are "no." If either faithfulness or factual hallucination is "yes," we treat the sentence as factual. To construct summary level labels, we first apply this rule to every sentence and then group sentences by their original FareBio summary. If a summary contains at least one sentence that is not factual, we label the whole summary as not factual, which produces 33 summaries. If all sentences in a summary are factual, we label the summary as factual, which produces 174 summaries. For the subset of 53 valid summaries used in our explanation focused evaluation, we fur-

[18]Can be downloaded here: `https://github.com/salesforce/QAFactEval/blob/master/download_models.sh`

[19]`https://www.nltk.org/`

[20]Download here: `https://github.com/salesforce/QAFactEval/blob/master/download_models.sh`

ther keep only those sentences that meet our filtering criteria for factual elaborative explanations and then aggregate them back into summaries.

For the computational costs of `GPT-4o`, it costs about $0.01 per plain language summary during the factual consistency evaluation on the CELLS dataset experiment, combining input and output tokens, and takes about 2 seconds per summary. In contrast, PLAINQAFACT takes about 30 seconds per summary on our hardware. However, PLAINQAFACT is able to provide intermediate outputs for each step in the QA based evaluation, which makes our metric more transparent and easier to inspect than the black box `GPT-4o` evaluation, even in most cases where `GPT-4o` achieves slightly higher performance than PLAINQAFACT.

Appendix E.4. LLM Prompts

We utilize two types of prompts to guide LLMs through two stages of factual consistency evaluation: classification and answer extraction. In accordance with the benchmark annotation protocol (Appendix A), we employ `GPT-4o` as a classifier to determine whether a given sentence or summary requires elaborative explanations for factual verification. For both stages, we set the `max_tokens` parameter to 512 and configure the temperature to 0 for `GPT-4o` and 0.01 for the `Llama 3.1 8B Instruct` model.

Developer

```
Annotate whether a sentence or summary includes information not present in the original abstract.

The sentence or summary contains external information that is not explicitly mentioned, paraphrased,
or implied in the original abstract will be labeled as 'Yes'.

The sentence or summary contains information that is explicitly stated or closely paraphrased from
the original abstract will be labeled as 'No'.
```

User

```
Sentence or summary: <input>
Original abstract: <abstract>
```

Example 1: Prompt of `GPT-4o` as the Classifier.

For the AE stage, we explore both `GPT-4o` and `Llama 3.1 8B Instruct` as backbone models to assess the factual consistency evaluation performance of PLAINQAFACT. Following the task description outlined in QAFactEval [39], we instruct both LLMs to extract potential answer entities from the input PLS.

Developer

```
QA-based metrics compare information units between the summary and source, so it is thus necessary to
 first extract such units, or answers, from the given summary. Please extract answers or information
units from a plain language summary.
```

User

```
Extract a comma-separated list of the most important keywords from the following text: <input>
```

Example 2: Prompt of `GPT-4o` as the Answer Extractor.

```
System
QA-based metrics compare information units between the summary and source, so it is thus necessary to
 first extract such units, or answers, from the given summary. Please extract answers or information
units from a plain language summary.

User
Extract a comma-separated list of the most important keywords from the following text: <input>
```

Example 3: Prompt of `Llama 3.1 8B Instruct` as the Answer Extractor.

Additionally, we report the performance of using `Llama 3.1 8B Instruct` as a judge to evaluate the factual consistency of a given PLS based on its source scientific abstract.

```
System
Rate the factuality of the given plain language sentence or summary compared with the scientific
abstract. Output a numeric score from 0 to 100, with 100 meaning the sentence is completely factually
 consistent with the abstract and 0 meaning the sentence is completely non-factual with the abstract.

User
Sentence or summary: <input>
Original abstract: <abstract>
Factuality score (only output a numeric score): <score>
```

Example 4: Prompt of `Llama 3.1 8B Instruct` as a factual consistency judge.

For the LLM-based perturbation of PLAINFACT and CELLS datasets, we follow the protocol of APPLS [12] on faithfulness criteria.

```
System
You are a data transformation assistant. You will receive a sentence from a biomedical literature.
You will generate a new version of the given sentence based on the following rules for faithfulness
perturbations:

1. Number Swap
Locate any numeric value(s) in the sentence and swap them with different numeric value(s).
Example: "infected more than 59 million people" -> "infected more than 64 million people"

2. Entity Swap
Locate a key entity (e.g., virus name, drug name, organization) in the sentence and swap it with a
different entity.
Example: "coronavirus 2 (SARS-CoV-2)" -> "canine adenovirus (CAV-2)"

3. Synonym Verb Swap
Identify a key verb in the sentence and replace it with a near-synonym or related verb that changes
the nuance or meaning slightly.
Example: "killed more than one of them" -> "stamped out more than one of them"

4. Hypernym/Antonym Swap
Select a word and replace it with either a hypernym (a more general term) or an antonym (opposite
meaning), as appropriate.
```

```
Example (antonym): "killed more than one of them" -> "saved more than one of them"
Example (hypernym): "dog" -> "animal" (if relevant)

5. Negation
Negate a key part of the sentence to flip its meaning.
Example: "has infected more than 59 million people" -> "hasn't infected more than 59 million people"
or "has not infected more than 59 million people"

Your task:
Read each sentence, generate a new sentence based on one of the five types of perturbation stratigies
 (Number Swap, Entity Swap, Synonym Verb Swap, Hypernym/Antonym Swap, Negation) above.
Return the perturbation sentence.

Do not change the rest parts of the sentence except for the perturbation content. For example, the
original sentence is "The skin patch and the vaginal (birth canal) ring are two methods of birth
control." The perturbation sentence should be "The skin patch and the vaginal (birth canal) ring are
five methods of birth control."

User
Sentence: <input>
Perturbation sentence: <sentence>
```

Example 5: Prompt of `GPT-4o` for faithfulness perturbation

Appendix F. Pilot Study on FactPICO

To investigate the performance of existing automatic factual consistency evaluation metrics on plain language generation tasks, we employ the recently introduced FactPICO dataset [9]. This dataset comprises human-labeled plain language summaries of Randomized Controlled Trials (RCTs) that address several critical elements: Populations, Interventions, Comparators, Outcomes (PICO), and additional information. All summaries are generated by various LLMs based on medical literature and include added information (i.e., extensive explanations) not present in the original abstracts. We hypothesize that existing factuality evaluation metrics for text summarization may struggle to accurately assess the factuality of this added information. To validate this assumption, we conduct pilot studies using four factual consistency evaluation metrics from FactPICO alongside one NLI-based metric: DAE [48], AlignScore [19], SummaC [16], QAFactEval [39], and QuestEval [50].

Appendix F.1. Dataset Pre-processing

FactPICO provides span-level annotations for LLM-generated summaries, assessing whether the additional information is present and determining its factuality, labeled as either "yes" or "no." We first remove all special identification tags from the abstracts, such as "ABSTRACT.BACKGROUND," "ABSTRACT.RESULTS," and "ABSTRACT.CONCLUSIONS." We then deduplicate the generated summaries,

while retaining duplicates within abstracts, as each abstract has three summaries generated by different LLMs, and each abstract may have varying numbers of generated summaries, from none to multiple. This pilot study focuses on two key research questions: evaluating the effectiveness of existing factuality metrics in assessing added information and non-factual added information.

Appendix F.2. Experiments and Results

We conduct experiments using the processed FactPICO dataset to conduct pilot studies that assess the performance of existing factual consistency evaluation metrics in detecting added information in plain language summarization tasks. It is important to note that some outliers in the plain language summaries of the FactPICO dataset contain extraneous content, which causes the input lengths to exceed the limitations of both the DAE and QAFactEval metrics. Furthermore, the factuality of the LLM-generated plain language summaries is not guaranteed.

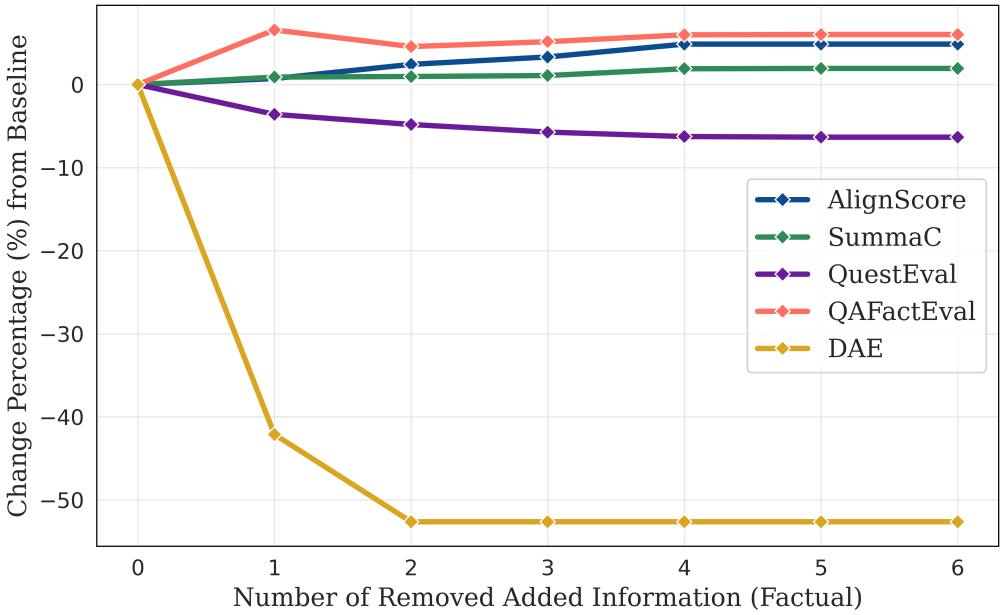


Figure F.3: Score change percentage from baselines over five metrics on the FactPICO dataset in removing factual added information. We expect each metric stays unchanged even when more added factual information is removed. The evaluation dataset contains 88 valid summary-abstract pairs.

RQ1: Do existing metrics perform well when external information is added to plain language summaries compared with no added information?

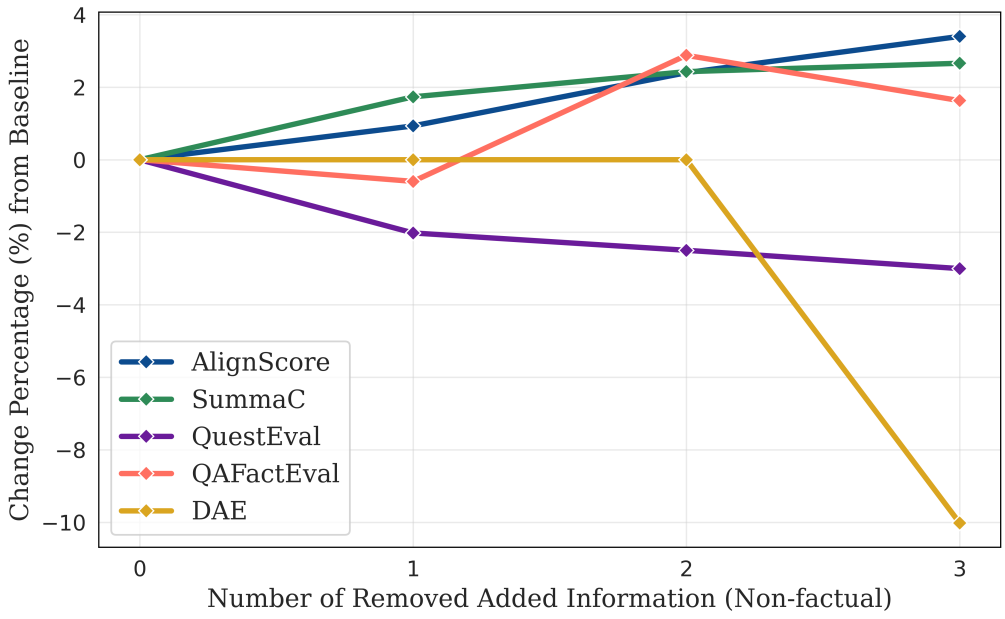


Figure F.4: Score change percentage from baselines over five metrics on the FactPICO dataset in removing non-factual added information (60 pairs). We expect the change percentage from baseline increases when more added non-factual information is removed.

In this study, we evaluate the ability of existing factual consistency evaluation metrics to detect added information in plain language summaries. We focus on summaries where all added information is annotated as factual, resulting in a dataset of 88 summary-abstract pairs. To assess metric sensitivity to the added information, we iteratively remove sentences from each plain language summary that contain added spans. For example, if a span such as "of a medicine called haloperidol" is labeled as factual ("yes"), we remove the entire sentence in the original plain language summaries containing that span through exact matching, continuing this process until no added information remains. This procedure enables us to determine how effectively current metrics handle added information that is absent from the original abstracts. As shown in Figure F.3, we report performance changes relative to baseline scores. For example, the AlignScore evaluation increases by 3.3% (on a 100-point scale) when six spans of added information are removed, resulting in an overall change of approximately 4.9% compared to the setting in which no added information is removed. These findings indicate that added information affects model-based factuality evaluation metrics such as QAFactEval (6.0%), AlignScore (4.9%), and SummaC (1.9%), with scores improving as more added information is removed. In contrast, both

QuestEval and DAE scores decline with the removal of added information, and notably, DAE exhibits the most significant performance drop. Overall, these findings suggest that all the five metrics have difficulty accurately assessing added factual information, as evidenced by both increases and decreases in their performance.

RQ2: Can existing metrics distinguish between non-factual and factual added information?

In this research question, our goal is to evaluate the sensitivity of factuality evaluation metrics in detecting non-factual information within plain language summaries. The FactPICO dataset labels added information as either "yes" (factual) or "no" (non-factual). In this scenario, we sample only those plain language summaries that contain both "yes" and "no" labels for added spans. As with RQ1, we iteratively remove sentences containing non-factual added spans from each plain language summary until no non-factual sentences remain. Overall, as illustrated in Figure F.4, only AlignScore, SummaC, and QAFactEval show improved performance as more non-factual information is removed, indicating that these metrics are sensitive in added non-factual information. Nevertheless, based on the results of RQ 1 and 2, our findings suggest that existing factual consistency evaluation metrics have limited capacity to accurately distinguish between factual and non-factual added information in plain language summarization tasks.

Appendix G. Statistical Testing

Following SummaC [16] and QAFactEval [39], we test whether our proposed metric achieves statistically significant improvements over other methods. Given Kendall's τ and Pearson correlations do not show the discriminative feature of continuous scores in various evaluation metric, we run a systematic evaluation on four datasets (CELLS, FareBio, FactPICO, and PLAINFACT) through bootstrap resampling [59] on the AUC-ROC results. We compare our best metric to other methods using the confidence intervals as significance level of 0.01 and apply Bonferroni correction [60] following SummaC [16]. Our results show that PLAINQAFACT achieves statistically significant improvements over QuestEval [50], SummaC [16], and DAE [48] on the CELLS and PLAINFACT datasets. As shown in Figure 2, PLAINQAFACT also outperforms other automatic metrics in evaluating elaborative explanation sentences, except for QAFactEval [39]. This highlights the effectiveness of PLAINQAFACT for biomedical PLS tasks that include external explanations.

For hypothesis testing, we treat AUC-ROC as the primary metric. For each baseline b we conduct a paired bootstrap test on the difference in AUC-ROC, $\Delta^{(k)} = \mathrm{AUC}^{(k)}_{\textsc{PlainQAFact}} - \mathrm{AUC}^{(k)}_{b}$ for bootstrap replicate k. The one-sided p-value for the alternative "PLAINQAFACT > baseline" is estimated as $p = \Pr(\Delta \leq 0)$ over the 10,000

replicates. Across the seven baselines we apply Holm-Bonferroni [52] correction to control the family-wise error rate at $\alpha = 0.01$.

Appendix H. Entity Swap Perturbation

Dataset	Metrics	Kendall's τ	Pearson	AUC (99% CI)
CELLS	Llama 3.1	24.6	29.9	65.7* [57.8, 73.3]
	GPT-4o	69.3	74.3	98.4 [96.4, 99.7]
	QAFactEval	55.9	64.6	89.4 [85.5, 93.2]
	QuestEval	22.6	27.0	66.0* [58.7, 72.8]
	SummaC	16.4	24.5	61.7* [54.2, 68.7]
	AlignScore	36.7	44.6	75.9* [69.5, 81.8]
	DAE	20.6	20.6	56.8* [52.6, 60.9]
	PlainQAFact	56.9	68.2	90.2 [85.9, 93.9]

Table H.9: Evaluation results of automatic metrics on the CELLS dataset [28] with entity swap non-factual plain language summary perturbation. Results are evaluated using Kendall's τ, Pearson correlation, and AUC (AUC-ROC). * indicates a statistically significant difference compared to PlainQAFact ($p < 0.01$). The standard deviations (std.) over five runs are: 0.1 (PlainQAFact), 0.2 (`Llama 3.1`), and 2.5 (`GPT-4o`).

Dataset: CELLS	AE	Kendall's τ	Pearson	AUC (99% CI)	std.
w/ Heuristic Entity Swap	Llama 3.1	**56.9**	**68.2**	**90.2** [85.9, 93.9]	0.1
	PromptRank	49.6	58.8	85.0* [79.7, 89.6]	-
w/ LLM Perturbation	Llama 3.1	**62.1**	**75.2**	**93.8** [90.3, 96.7]	0.1
	PromptRank	56.3	66.6	89.6* [85.3, 93.3]	-

Table H.10: Ablation study of answer extraction approach in the PlainQAFact on the CELLS dataset. w/ Heuristic Entity Swap uses APPLS entity swap [12] approach to create non-factual plain language summaries, while w/ LLM Perturbation applies `GPT-4o` to perturb the plain language summaries. AE = answer extraction. AUC = AUC-ROC. * indicates a statistically significant difference compared to `Llama 3.1` ($p < 0.01$). We run each setting for five times and report the standard deviation (std.). Note that only the settings using LLMs may produce fluctuated scores.

As we discussed in Section 5.1, in addition to evaluating the performance of automatic factual consistency evaluation metrics on LLM-perturbed datasets, we also use the non-LLM entity swap approach introduced in Guo et al. [12] to assess the robustness of our proposed metric. Specifically, we perturb plain language summaries using the CELLS dataset following the same entity swap protocol as APPLS [12].

The entity swap system identifies specific "entity spans" in the text and substitutes them with existing terms from the UMLS knowledge base. As shown in Table H.9, PLAINQAFACT achieves the best performance among all automatic evaluation metrics on the CELLS dataset. PLAINQAFACT achieves the highest correlation with human judgement (AUC=90.2, 99% CI [85.9, 93.9]). As comparative effect sizes, we report pairwise differences between PLAINQAFACT and the strongest baseline QAFactEval. The ΔAUC = 0.7, where Δ is PLAINQAFACT minus the baseline. After correction, PLAINQAFACT shows significantly higher AUC than `Llama 3.1`, QuestEval, SummaC, AlignScore, and DAE ($p < 0.01$, Holm-Bonferroni corrected for all), but is not significantly different from QAFactEval ($p = 0.64$) or `GPT-4o` ($p = 1.0$).

We also report an ablation study using a non-LLM answer extractor to show the effectiveness of our LLM-powered answer extraction approach. In Table H.10, our answer extraction method using `Llama 3.1` consistently achieves higher correlation (AUC=90.2, 99% CI [85.9, 93.9] in w/ heuristic entity swap and AUC=93.8, 99% CI [90.3, 96.7] in w/ LLM perturbation) compared with the non-LLM answer extractor (i.e., PromptRank). Using `Llama 3.1` shows significantly higher AUC than PromptRank in both settings ($p < 0.01$, Holm-Bonferroni corrected for all), highlighting the robustness and effectiveness of LLM-powered answer extraction in factual consistency evaluation.

Appendix I. Summary-Level Explanation-Only Evaluation

Similar to our sentence-level explanation annotations in PLAINFACT, FactPICO [9] and FareBio [47] datasets also provide human-annotated explanation information in plain language summaries. According to the results shown in Section 5.2, we also assess the performance of five factual consistency evaluation metrics under FactPICO and FareBio datasets.

In the FactPICO, the elaborative explanation is defined as "added information." We first select summaries that include added information, and then we label a summary as "non-factual" if it contains any non-factual added information as determined by annotators. We generate a summary-level explanation-only FactPICO dataset consisting of 231 summary-abstract pairs. As shown in Figure I.5, AlignScore achieves the best performance across all three evaluation criteria. However, since the summaries in the FactPICO dataset are generated using LLMs, the factual consistency between the generated summaries and the original source abstracts is not guaranteed. For instance, we treat a summary as "factual" indicates that all the added information is factual, but it does not necessarily reflect the factual consis-

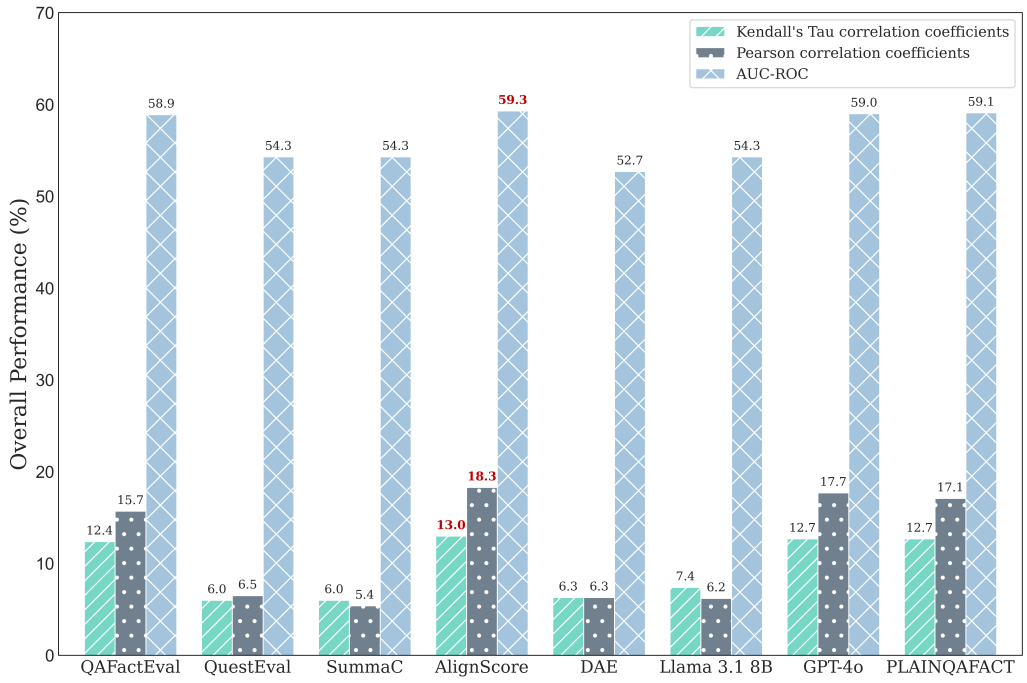


Figure I.5: Overall performance on summaries containing added information (i.e., elaborative explanations) from FactPICO [9]. The std. of PLAINQAFACT, `Llama 3.1`, and `GPT-4o` are 0.2, 0.2, and 3.3, respectively, based on five runs of each metric.

tency of other content. Therefore, the results presented in Figure I.5 may become biased.

For FareBio [47], we collect explanation sentences based on both "faithfulness" and "factual hallucination" labels in the original dataset. As shown in Figure I.6, PLAINQAFACT achieves the best performance compared with all other metrics, showing a clear advantage over `GPT-4o`. Since the FareBio dataset provides sentence-level annotations of factual consistency for each sentence in the plain language summaries, the results are more reliable than those from FactPICO. On both datasets, we do not find statistically significant differences in AUC-ROC metric by $p < 0.01$. While some existing metrics perform similarly in certain settings, our metric consistently shows better performance in correlation.

Appendix J. Quantitative Error Analysis of FareBio

Given the Table 6, we further conduct quantitative analysis regarding failure cases in FareBio using our proposed metric. We randomly sample 30 plain language

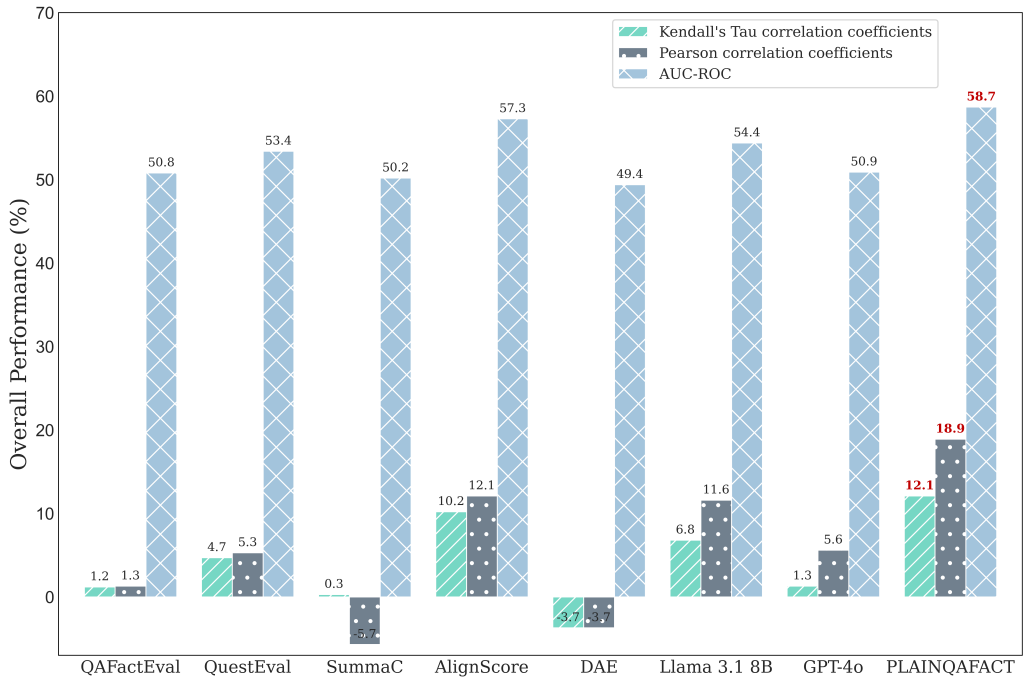


Figure I.6: Overall performance on summaries with elaborative explanations from FareBio [47]. The std. for PLAINQAFACT, `Llama 3.1`, and `GPT-4o` are 0.03, 0.3, and 5.7, respectively, computed over five runs.

summaries (both factual and non-factual) from FareBio and evaluate all sentences within them, resulting in 324 split sentences. We categorize errors into three main types:

1. QA Pipeline Errors (34.0% of errors): misclassification of sentence types, retrieval of irrelevant context, and generation of unanswerable questions.

2. Answer Extraction and Evaluation Issues (18.4% of errors): failure to extract numerical values and answer mismatches when using BERTScore.

3. Dataset Labeling Issues (47.7% of errors): inconsistencies in FaithBio's annotation protocol.

Table J.11 presents the complete quantitative breakdown. We identify 109 errors across the 324 sentences (33.6% error rate). Notably, 47.7% of errors occurs in sentences mislabeled as factual and non-factual, with most being false negatives. FareBio uses a different annotation protocol than PLAINFACT. All source-grounded simplification sentences are implicitly treated as factual. Annotators do not evaluate factuality once a sentence is labeled as faithful. This creates label noise when evaluating simplification sentences, as the ground truth may not reflect actual factual

consistency. Our analysis reveals that PLAINQAFACT performs better on PLAINFACT (where annotation protocols align with our metric's design) than on FareBio. The performance gap reflects both limitations in our approach, particularly for numerical verification, and incompatibilities between FareBio's annotation protocol and our QA components.

Error Type	Count	% of Errors	% of Total	In F	In NF
QA Pipeline Errors					
Misclassification	13	11.9%	4.0%	10	3
Retrieval quality	9	8.3%	2.8%	8	1
Unanswerable generated question	15	13.8%	4.6%	11	4
Answer Extraction and Evaluation Issues					
Numerical value mis-extraction	16	14.7%	4.9%	11	5
Answer match misalignment	4	3.7%	1.2%	3	1
Dataset Issues					
False positive (F labeled as NF)	10	9.2%	3.1%	0	10
False negative (NF labeled as F)	42	38.5%	13.0%	42	0
Total Errors	109	100.0%	33.6%	85	24
Correct Classifications	215	–	66.4%	114	101
Total Sentences Analyzed	324	–	100.0%	198	126

Table J.11: Quantitative error analysis of PLAINQAFACT on 60 randomly sampled summaries from FareBio dataset (30 factual, 30 non-factual). Summaries are split into sentences when evaluating the factual consistency. F = factual sentences; NF = non-factual sentences.

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